Korean-Centered Cross-Lingual Parallel Sentence Corpus Construction Experiment

Danxin Cui
College of Foreign Language and Literature
Fudan Univeristy
Shanghai, China
22110120003@m.fudan.edu.cn

Yude Bi*
College of Foreign Languages and Literature
Fudan University
Shanghai, China
biyude@fudan.edu.cn

***Abstract*—This paper proposes an efficient method for parallel sentence extraction based on LASER model. For our work, we mined reports in 4 languages from DONG-A ILBO website dating from 1999 to 2022. We designed custom sentence segmentation schema based on language features, and regulate the quality of selected sentence with suitable threshold. Experiment on a relatively small data of 2,000 articles (500 articles for each language) demonstrated the outstanding performance of our method, which has laid a solid foundation for our construction of large diachronic Korean-centered four-language parallel sentence corpus.**

Keywords— Korean-centered four-language parallel sentence corpus, Cross-Lingual embeddings, semantic similarity

I. Introduction

Cross-lingual parallel corpora stand as indispensable resources for researches in various disciplines in the present era. In the realm of NLP, they are necessary for machine translation model training and evaluation[1], and cross-lingual transfer learning[2], contribute to the advancement of language understanding and generation tasks and enable the development of models that can retrieve relevant information across multiple languages[3]. In the field of linguistics, they enable comparative analyses of languages, dialects, and language evolution, providing insights into language structure, semantics, and usage patterns. Moreover, specialized multilingual corpora cater to high-quality data resources within specific domains. However, in contrast to monolingual corpora, parallel corpora's sources cover a restricted range of text types such as legislation, administration, and technical documentation. Furthermore, they are limited to high-density languages such as English and the official European languages[4].There are only few parallel corpora containing Korean. Moreover, the existing Korean-related multilingual parallel corpora often face the following issues: small data volumes, limited text-type coverage, short time spans and outdated content.

Therefore, we aim to utilize the news reports from the Dong-A ILBO website spanning from 1999 to 2022, covering seven different domains, as the primary data source. The goal is to construct a large-scale diachronic multilingual parallel corpus with Korean as the central language. This endeavor seeks to enable Korean to get the full benefit of NLP, ensuring high-quality data resources for multilingual teaching and comparative research involving the Korean language. Additionally, it intends to offer high-quality data resources for researches in the field of journalism.

The rest of the paper is organized as follows; Section 2 outlines the related work. Section 3 describes the parallel sentence extraction method. Section 4 presents the result evaluations and offers discussions, and section 5 gives our conclusions.

*Yude Bi is the corresponding author.

II. Related Work

A. Korean-involved Multilingual Parallel Sentence Corpora

TABLE I presents existing multilingual parallel corpora containing Korean data. The KAIST Translation Evaluation Set and KAIST Chinese-Korean Multilingual Corpus are developed by the Korea Advanced Institute of Science and Technology (KAIST) to support research in Korean text processing. The former is primarily employed for testing English-Korean machine translation tasks, while the latter encompasses 60,000 pairs of Chinese-Korean short sentence pairs, predominantly catering to Chinese-Korean machine translation tasks. PAW-X comprises 23,659 pairs of manually translated PAWS (Paraphrase Adversaries from Word Scrambling) evaluation sentences and approximately 300,000 pairs of machine-translated sentences, encompassing six languages, including Korean. Among these, Korean accounts for approximately 5,000 training pairs, 1,965 validation pairs, and 1,972 test pairs. Sae4k includes about 50,000 pairs of utterances for questions and instructions, paired with natural language queries. The Korean Parallel Corpora encompass roughly 100,000 English-Korean sentence pairs collected from news sources, primarily utilized for English-Korean machine translation purposes.

Previous Korean-related parallel corpora contributed greatly to Korean natural language processing academic extension. However, we have found that there is a lack of Korean-centered multilingual corpora and the existing Korean-related corpora are outdated and limited in scope. Therefore, we plan to construct a cross-lingual corpus using data from the Dong-A ILBO website. This approach allows us to utilize the vast amount of existing data on the Dong-A ILBO website while also taking advantage of the real-time updates in news articles to ensure the timeliness of the corpus.

B. Sentence Alignment

Text Alignment, at its core, is the process of establishing correspondence between two text data entities. In practical applications, researchers typically break down the data entities to be compared into finer-grained units, express them in a consistent format, and ultimately employ scoring functions to identify and align relevant units.

In this section, we will provide a brief overview of three methods for parallel text alignment: length-based[5], translation-based and embedding-based methods. The length-based involves comparing the lengths of the source and target sentences. Translation-based methods rely on the vocabulary

2023 International Conference on Asian Language Processing (IALP) | 979-8-3503-3078-6/23/$31.00 ©2023 IEEE | DOI: 10.1109/IALP61005.2023.10336988

TABLE I PARALLEL SENTENCE CORPORA CONTAINING KOREAN

Korean-related Parallel Corpus	Languages	Korean Data Volume	Date
KAIST Translation Evaluation Set	Ko → En	3000 pairs	2001
KAIST Chinese-Korean Multilingual Corpus	Ko → Zh	60,000 pairs	2005
PAW-X[6]	En → {Fr, Es, De, Zh, Ja, Ko}	8937 sentence pairs	2019
Sae-4k[7]	Ko → En	50,837 pairs	2020
Korean parallel corpora[8]	Ko → {En, Fr}	100k sentence pair	2020

correspondence between target and source sentences[9]. Embedding-based methods entail transforming text data into numerical representations in a high-dimensional space. For example, the LASER model[10] employs BiLSTM as an encoder and LSTM as a decoder, and it was trained on a dataset comprising 223 million sentence pairs across languages. Its success has laid a solid foundation for the development of large-scale multi-lingual sentence representation.

III. Method

Our parallel corpus contains parallel sentences between Korean (ko) and 3 other languages, that is, Chinese (ch), English (en) and Japanese (ja). Furthermore, it also comprises parallel sentences between the other 3 language pairs acquired by pivoting through Korean (ko). As detailed in this chapter, we first aggregated all accessible reports in the 4 languages that have been published on the website of DONG-A ILBO from 2000 to 2022. Then we had them segmented by sentences, embedded sentences with LASER model and find parallel sentence pairs according to the Euclidean distance of their embeddings.

A. Collation from Accessible Reports

The website of DONG-A ILBO publishes reports of the same content in Korean, Chinese, English and Japanese, which makes it a natural source for parallel corpus. We build custom extractor using BeautifulSoup to extract article and arrange them by years. The unique ID for articles of the same content but different languages is also kept. We ultimately collected a total of 264,173 articles across 22 years (2000-2022), covering seven sections: Sports, Politics, Business, Culture, National, International and Editorial.

B. Data Preprocessing

We opted against extensive text cleaning (removing stop words, special character and low-frequency word, lemmatization and tokenization, etc.) to avoid potential loss of semantic meaning. After pre-segmenting sentences using SpaCy (ko and en), Jio (ch) and Hasami (ja), we designed adaptive segmentation schema for different language based on their punctuation usage principles and writing style. For example, splits text by terminal punctuation. Whereas, Hasami treats sentences enclosed within the same quotation marks as a single sentence. This difference can lead to misalignment during the parallel sentence matching process. To address this issue, we devised sentence segmentation as shown in Algorithm 1.

C. Parallel Sentence Extraction

Employing segmentation schema mentioned above, we segmented 4 reports in different languages of the same content at the same time to search for parallel sentences. Let $S = \{s_1, s_2, \ldots, s_n\}$ be the set of all sentences in English article. Let $K = \{k_1, k_2, \ldots, k_n\}$ be the set of all sentences in Korean article. Let $J = \{j_1, j_2, \ldots, j_n\}$ be the set of all sentences in Japanese article. And let $C = \{c_1, c_2, \ldots, c_n\}$ be the set of all sentences in Chinese article. Next, we set the scoring function which assigns a score indicating the similarity between a Korean sentence embedding and sentence embeddings from the other 3 languages. Taking Korean sentence and English sentence as an example, we set the scoring function as $f(k, s)$, where $k \in K$ and $s \in S$. Given a Korean sentence $k_i \in K$, its corresponding English sentence can be found as:

$$s^* = \min_{s \in S} f(s, k_i) \quad (1)$$

We chose f to be the Euclidean distance function on embedding. And for embedding computation model, we opted LASER over LaBSE in order to eliminate the interference of word position information. Considering that Korean, Chinese and English have significant word order differences, which yet have little influence on semantic meaning (detailed explanation of their differences is given in Chapter IV). Next, we computed embeddings using LASER model that encodes text from different languages into a shared embedding space. The Euclidean distance between the LASER embeddings pairs is referred as the LASER Euclidean Alignment Score (LEAS).

In the process of alignment, it is notable that articles in different languages of the same content is not always a sentence-by-sentence translation of each other. Sometimes, 2 Korean sentences correspond to a single sentence in Chinese/English/Japanese, or two Chinese/English/Japanese sentences match a single Korean sentence. Such examples are presented in TABLE II. There are also instances where sentences appear in Korean articles but not in the other language articles, and vice versa. In response to these scenarios, taking ko-ja pairs as an example, apart from calculating the LEAS between sentence pairs (k_i, j_i), we also measured the LEAS between pairs (k_i, j_i+j_{i+1}) and (k_i+k_{i+1}, j_i) to tackle the situation where one sentence corresponds to two sentences. Additionally, we determined the LEAS of sentence pair (k_{i+1}, j_i) and of sentence pair (k_i, j_{i+1}) for sentence skipping issue. Subsequently, the LEAS of these five pairs of sentences were compared and sentence pair of the smallest LEAS would be incorporated into the parallel corpus along with its LEAS score.

Then a parallel sentence extraction experiment based on the method mentioned above was carried out on a dataset comprising 200 articles of 4 languages. The result demonstrated that for ko-ja sentence pairs, a LEAS below 0.46 is required to guarantee semantic similarity between the extracted sentences. The thresholds for ko-ch pairs and ko-en

TABLE II INSTANCES WHERE 2 SENTENCES IN A CERTAIN LANGUAGE CORRESPOND TO 1 SENTENCE IN ANOTHER LANGUAGE, TAKING KOREAN-ENGLISH AS AN EXAMPLE

	Korean	English
2 Korean sentences vs. 1 English sentence	박 전 대통령은 앞으로도 최소 한 달은 병원에서 치료에 전념할 것으로 알려졌다. 어깨 질환과 허리 디스크 등 지병과 치과, 정신건강의학과 등 치료를 받고있다.	She is expected to stay in hospital for at least one month until she recovers from a shoulder condition, a herniated lumbar disc, dental issues and mental sickness.
1 Korean sentences vs. 2 English sentences	그는 북한은 자신들이 취한 행동이 도발이라기보다는 안보 위협에 대처한 일종의 결의 또는 전쟁 발발 가능성에 대비한 준비 과정으로 생각하고 있었다고 말했다.스트롱 특사는 이라크전쟁은 북한 핵 위기의 평화적 해결을 위한 특별한 동기가 될 수 있을 것이라며 그러나 현실적 충격이 어떤 가능성을 갖고 올지에 대해 단언하기는 어렵다고 말했다.	He said, “It seemed to consider their actions not a provocation but a kind of resolution to deal with the threat on its security or preparatory movement for the possible war. Iraq war can be a good opportunity for peaceful resolution of the issue involving North Korea`s nuclear development. But it is hard to predict the actual result for now.”

Algorithm 1 Sentence segementation algorithm for Korean

Input： Sentences segmented by SpaCy $[O_1, O_2, \ldots, O_i]$

Output： Sentences after adjustment $[N_1, N_2, \ldots, N_m]$

```
old_sents ← [O1,O2,…,Oi]
indices_start_quote ← [indexs of all sentences that contains front quotation mark]
indices_end_quote ← [indexs of all sentences that contains end quotation mark]
indices_interlock ← [] //record indexes of sentences within quotation marks
indices_interlock_new ← []
for (i = 0 to len(indices_start_quote)-1) do
  append (indices_start_quote[i], indices_end_quote[i]) to indices_interlock
for (i = 0 to len(indices_interlock)-1]) do
  if len(indices_interlock_new) == 0 then
    append indices_interlock[i] to indices_interlock_new
  else
    if indices_interlock[i][0] == indices_interlock_new[-1][1] then
      Replace the last element of indices_interlock_new with
       (indices_interlock_new[-1][0], indices_interlock[i][1])
    else
      append indices_interlock[i] to indices_interlock_new
all_range ← [i for (i = 0 to len(old_sents))]
for (g = indices_interlock_new[0] to indices_interlock_new[-1]) do
    quote_range ← [n for (n =g[0] to g[1])]
  all_range ← The set difference between all_range and quote_range //indexes of sentences outside quotation marks
    sent_to_sep ← put sentences whose index is in list quote_range together
    seped_sents ← split sent_to_sep at the end mark right after the end quotation mark
   //Predicates in Korean are usually at the end of sentences
new_sents ← put sentences in seped_sents with sentences outside of quotations marks together according to their original order
return new_sents
```

pairs are 0.50 and 0.56, respectively. Therefore, we applied a threshold and select only those pairs for which the LEAS is smaller than the threshold.

IV. EVALUATION RESULTS AND DISCUSSION

We experimented our method on a small dataset of 2000 article (500 for every language) and successfully constructed a cross-lingual parallel sentence corpus. In the following parts, we will evaluate the quality of this corpus from 3 perspectives: data utilization rate, LaBSE evaluation and annotation evaluation.

A. Data Utilization Rate and Discussion

An optimal extraction method should select sentence pairs meeting the stipulated similarity criteria under the premise of a satisfactory utilization rate of the original data. We have set thresholds to guarantee the quality of our parallel corpus, which inevitably came at the expense of data loss. In order to evaluate our data utilization rate, we counted the total number of sentences in the original data and the number of parallel sentence pairs for the ko-ch, ko-en and ko-ja language pairs, respectively. In practice, the ratio of the number of parallel sentence pairs to the sentence count of sentences of the language with smaller number of sentences in the language pair represents the data utilization rate.

TABLE III UTILIZATION RATE OF ORIGINAL SOURCES

Language pairs	Sentence	Bitext pair	Utilization rate
en ko	5998 5292	4480	84.6%
ch ko	5529 5292	4858	91.7%
ja ko	5142 5292	4898	95.2%

As illustrated in TABLE III, ko-ja pair is with the highest data utilization rate (95.2%), the ko-ch pair, whose data utilization rate is 91.7%, follows slightly behind. While the data utilization rate of ko-en pair is far beneath the first two, which is only 84.6%. This is because:

- Both Korean and Japanese are agglutinative languages, sharing high linguistic similarity and similar sentence segmentation habits. The difference in the total number of sentences is also the smallest among the three language pairs.
- Compared to the other two langugage pairs, the ko-en pair doesn't always involve sentence-by-sentence translation. English article is more of an adaptation than a translation to its Korean correspondant. Hence, there are instances where sentence in English article have no corresponding counterparts in Korean article, which leads to a relatively low data utilization rate of ko-en pair.

Overall, the data utilization rate is rather satisfactory. There is no significant loss of original data during the parallel sentence extraction process.

B. *Evaluation with LaBSE Model and Discussion*

LaBSE (Language agnostic BERT Sentence Embeddings) is a cross-lingual sentence embedding model supporting 109 languages developed by Google Research team in 2020. It can map sentences from different languages into a shared semantic space to achieve language agnostic sentence representations[11]. The model is based on BERT, and its training method combines methods for learning monolingual and cross-lingual representation: masked language modelling (MLM), translation language modelling (TLM)[12], dual encoder translation ranking[13] and additive margin softmax[14]. Pre-trained on 17 billion monolingual sentence and 6 billion bilingual sentence pairs, it demonstrated remarkable performance on several bi-text retrieval/mining tasks, and outperformed LASER on low-resource languages.

In view of its outstanding performance, we introduced LaBSE model for evaluation. To contrast with Euclidean distance, we also employed cosine similarity as a dual verification of semantic similarity. Cosine similarity calculates the cosine value of the angle between vectors, with values ranging from -1 to 1. When its absolute value of the cosine similarity is closer to 1, the angle between the vector is smaller, indicating a higher degree of similarity.

We sent extracted parallel sentence to LaBSE model, and calculated the Euclidean distance (LaBSE_eud) and cosine similarity (LaBSE_cosine) between the generated sentence embeddings. Then we compared them with the Euclidean distance and cosine similarity between sentence embeddings generated by LASER model, which is referred to as LASER_eud and LASER_cosine, for evaluation purpose.

Fig. 1, Fig. 2 and Fig. 3 depict the probability distribution of the four similarity measurements (LASER_eud, LASER_cosine, LaBSE_eud and LaBSE_cosine) of the three language pairs, respectively. It can be observed from Fig. 1 that the LASER_eud score for ko-ch parallel sentence pairs are mostly concentrated around 0.4, while their LASER_cosine score are centered around 0.9. As witnessed in Fig. 2, the LASER_eud of ko-ja parallel sentence pairs are distributed around 0.3, and their LASER_cosine score is predominantly above 0.9. Fig. 3 displays that the LASER_eud scores for ko-en parallel sentence pairs are focused around 0.5, with the portion below 0.5 slightly larger than the portion above 0.5. The LASER_cosine scores for these pairs are centered around 0.89. Data above illustrate that the sentence pairs in our parallel corpus exhibit a high degree of semantic similarity.

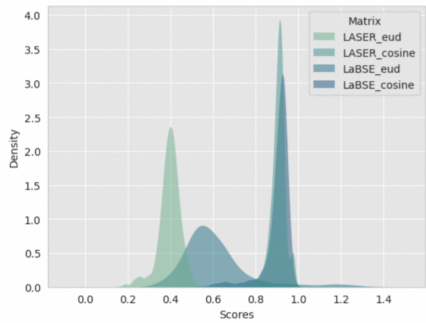

Fig. 1. Density plot of sentence similarity scores of ko-ch pair

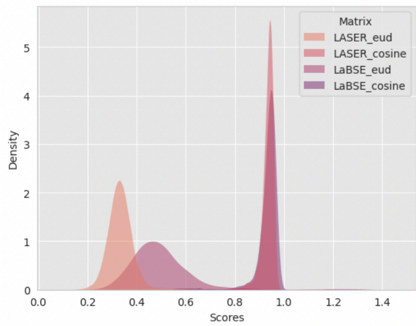

Fig. 2. Density plot of sentence similarity scores of ko-ja pair

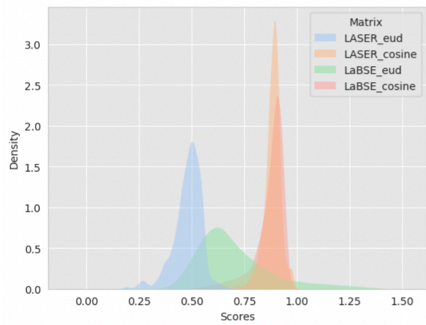

Fig. 3. Density plot of sentence similarity scores of ko-en pair

TABLE IV RESULTS OF THE ANNOTATION TASK TO EVALUATE THE SEMANTIC SIMILARITY BETWEEN SENTENCE PAIRS ACROSS 3 LANGUAGE PAIRS.

Language	Annotation data		STS score			Spearman correlation coefficient		
	Bitext pairs	Annotations	Definite accept	Marginal accept	Reject	LEAS, STS	LEAS, Sentence len	STS, Sentence len
ko-en	210	210	4.93	4.4	1.45	-0.86	0.45	-0.43
ko-ch	210	210	4.88	3.98	1.075	-0.83	-0.08	0.054
ko-ja	210	210	4.65	4.38	0.025	-0.77	0.32	-0.26

By comparing all data across the three graphs, we observed that the LaBSE_eud scores for all three language pairs are relatively higher contrast to their LASER_eud scores. This suggests that the distance between sentence vectors generated by the LaBSE model are higher than those generated by the LASER model. But it seems that this phenomenon is not adequate enough to substantiate the insufficiency of the quality of our corpus. Because the distribution ranges of LaBSE_cosine scores for the three language pairs are not significantly different from the LASER_cosine scores distribution ranges. Additionally, the LaBSE_cosine scores tend to be closer to 1, implying a high semantic similarity.

We speculate that this could be attributed to the different underlying architectures of the two models. LASER model takes the BiLSTM as its underlaying architecture, while LaBSE employs a Transformer embedding network. The Transformer architecture inherently considers positional information through positional encodings in its attention mechanism[15]. Whereas, the BiLSTM model processes input sequences in a sequential manner, considering only the context before and after each time step, without a clear way to capture the global positional information between two words. That is to say, the vectors generated by the BiLSTM model tends to have relatively less positional information of words compared to those generated by the Transformer model. Cosine similarity focuses on the relative differences between two vectors. On the other hand, Euclidean distance computes the absolute distance between vectors. When measuring the semantic similarity between sentences, compared to cosine similarity, Euclidean distance is more sensitive to the order and position of words within the text. As we know, different languages may lead to different word order, and LaBSE model is more capable of preserving positional information. Under this premise, the difference between LAER_eud and LaBSE_eud is naturally bigger than the difference between LASER_cosine and LaBSE_cosine.

C. Evaluation by Annotation

1）Annotation Setup

We randomly sampled 210 sentence pairs from the three language pairs. The sampling was stratified to have equal number of sentences from three set:

- *Definite accept*: sentence pairs with LEAS smaller than 0.1 of the chosen thresholds.
- *Marginal accept*: sentence pairs with LEAS smaller than but within 0.1 of the chosen thresholds.
- *Reject*: sentence pairs with LEAS bigger than but within 0.1 of the chosen thresholds.

The sampled sentence pairs were shuffled randomly such that no ordering is preserved across LEAS. We refer to the Sem-Eval-2016 Task 1[16] for defining the annotation score, wherein cross-lingual semantic textual similarity (STS) is defined by 6 ordinal levels ranging from complete semantic equivalence (5) to complete semantic dissimilarity (0). This guideline was explained to our annotators. For every language pair, we have assigned 1 annotator, who is fluent in both languages.

2) Annotation Results and Discussion

The results of the annotation of the 610 sentence pairs are shown language-wise in TABLE IV.

As can be observed, sentence pairs in our parallel corpus have high semantic similarity. According to the scoring guidelines followed by the annotators, a score of 5 represents the highest level of semantic similarity. The average STS scores for the “Definite accept” sentence pairs are 4.93 (ko-en), 4.88 (ko-ch) and 4.65 (ko-ja), all of which are close to 5. Even “Marginal accept” sentence pairs obtained relatively high STS scores. These statistics suggest that annotators rated sentence pairs in our corpus to be of high quality. Moreover, the chosen thresholds sensitively regulate quality: the “Definite accept” sentence pairs have a high average STS score of 4.82, which reduce to 4.25 with “Marginal accept” and significantly falls to 0.85 with the “Reject” sets.

For three language pairs, the Spearman correlation coefficients between LEAS and STS are strongly negative values of -0.86 (ko-en), -0.83 (ko-ch) and -0.77 (ko-ja), that is, sentence pairs with a lower LEAS are more likely to be rated as semantically similar. This proves the effectiveness of our parallel sentence extraction method and acknowledges the quality of our parallel corpus

We also examined the Spearman correlation coefficients between sentence length and LEAS as well as STS. To be consistent across languages, sentence length is set for the Korean sentence in every language pair. It is evident that the overall correlation between sentence length and both LEAS and STS is not strong. In the ko-en pair, sentence length shows a positive correlation with LEAS (shorter sentences are more likely to be judged as similar) and a negative correlation with STS (shorter sentences are more likely to be judged as similar). The situation in the ko-ja pair is similar to the ko-en pair, but the correlation between sentence length and LEAS/STS is weaker. In the ko-ch pair, the correlation coefficient between sentence length and both measures are close to 0, indicating that whether sentence pairs are judged as similar is not correlated with sentence length.

V. Conclusion and future work

In this paper, we have developed a system for cross-lingual parallel sentence corpus construction. We designed custom sentence segmentation method based on linguistic characters of different languages (Korean, Chinese, English and Japanese), employed LASER model for sentence embedding generation, and selected suitable LEAS thresholds to achieve dynamic parallel sentence searching.

We tested this system on a dataset of 2,000 reports from DONG-A ILBO website, and successfully built a high-quality four-language parallel sentence corpus centered around Korean language, which proved the effectiveness of our approach.

We are currently applying our system to a bigger dataset, which consists of 264,173 reports of 7 sections dating from 1999 to 2022, to build a diachronic Korean-centered four-language parallel sentence corpus. Once it is completed, we will publish it online as an open-source corpus to facilitate researches related to Korean language in the realm of NLP and linguistics.

References

[1] H. Sun, Y. Lei, and D. Xiong, "Multilingual Neural Machine Transliteration with Adaptive Segmentation Schemes," in *2022 International Conference on Asian Language Processing (IALP)*, Singapore, Singapore, 2022, pp. 494–499. doi: 10.1109/IALP57159.2022.9961282.

[2] A. Y. Maulana, I. Alfina, and K. Azizah, "Building Indonesian Dependency Parser Using Cross-lingual Transfer Learning," in *2022 International Conference on Asian Language Processing (IALP)*, Oct. 2022, pp. 488–493. doi: 10.1109/IALP57159.2022.9961296.

[3] X. Hu *et al.*, "Language Agnostic Multilingual Information Retrieval with Contrastive Learning," in *Findings of the Association for Computational Linguistics: ACL 2023*, Toronto, Canada: Association for Computational Linguistics, Jul. 2023, pp. 9133–9146. doi: 10.18653/v1/2023.findings-acl.581.

[4] I. Zeroual and A. Lakhouaja, "MulTed: a multilingual aligned and tagged parallel corpus," *Applied Computing and Informatics*, vol. 18, no. 1/2, pp. 61–73, Jan. 2020, doi: 10.1016/j.aci.2018.12.003.

[5] W. A. Gale and K. W. Church, "A Program for Aligning Sentences in Bilingual Corpora," *Computational Linguistics*, vol. 19, no. 1, pp. 75–102, 1993.

[6] Y. Yang, Y. Zhang, C. Tar, and J. Baldridge, "PAWS-X: A Cross-lingual Adversarial Dataset for Paraphrase Identification," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, Hong Kong, China: Association for Computational Linguistics, 2019, pp. 3685–3690. doi: 10.18653/v1/D19-1382.

[7] W. I. Cho, Y. Moon, S. Moon, S. M. Kim, and N. S. Kim, "Sae Machines Getting with the Program: Understanding Intent Arguments of Non-Canonical Directives," in *Findings of the Association for Computational Linguistics: EMNLP 2020*, Online: Association for Computational Linguistics, Nov. 2020, pp. 329–339. doi: 10.18653/v1/2020.findings-emnlp.31.

[8] J. Park, J.-P. Hong, and J.-W. Cha, "Korean Language Resources for Everyone," in *Proceedings of the 30th Pacific Asia Conference on Language, Information and Computation: Oral Papers*, Seoul, South Korea, Oct. 2016, pp. 49–58. Accessed: Aug. 23, 2023. [Online]. Available: https://aclanthology.org/Y16-2002

[9] F. Braune and A. Fraser, "Improved Unsupervised Sentence Alignment for Symmetrical and Asymmetrical Parallel Corpora," in *Coling 2010: Posters*, Beijing, China: Coling 2010 Organizing Committee, Aug. 2010, pp. 81–89. Accessed: Aug. 24, 2023. [Online]. Available: https://aclanthology.org/C10-2010

[10] M. Artetxe and H. Schwenk, "Massively Multilingual Sentence Embeddings for Zero-Shot Cross-Lingual Transfer and Beyond," *Transactions of the Association for Computational Linguistics*, vol. 7, pp. 597–610, Nov. 2019, doi: 10.1162/tacl_a_00288.

[11] F. Feng, Y. Yang, D. Cer, N. Arivazhagan, and W. Wang, "LaBSE Language-agnostic BERT Sentence Embedding." arXiv, Mar. 08, 2022. Accessed: Aug. 21, 2023. [Online]. Available: http://arxiv.org/abs/2007.01852

[12] A. CONNEAU and G. Lample, "Cross-lingual Language Model Pretraining," in *Advances in Neural Information Processing Systems*, Curran Associates, Inc., 2019, pp. 7059–7069. Accessed: Aug. 28, 2023. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2019/hash/c04c19c2c2474dbf5f7ac4372c5b9af1-Abstract.html

[13] M. Guo *et al.*, "Effective Parallel Corpus Mining using Bilingual Sentence Embeddings," in *Proceedings of the Third Conference on Machine Translation: Research Papers*, Brussels, Belgium: Association for Computational Linguistics, Oct. 2018, pp. 165–176. doi: 10.18653/v1/W18-6317.

[14] Y. Yang *et al.*, "Improving Multilingual Sentence Embedding using Bi-directional Dual Encoder with Additive Margin Softmax," in *IJCAI 2019*, Macao, China: arXiv, Aug. 2019, pp. 5370–5378. Accessed: Aug. 28, 2023. [Online]. Available: http://arxiv.org/abs/1902.08564

[15] A. Vaswani *et al.*, "Attention Is All You Need," *arXiv:1706.03762 [cs]*, Dec. 2017, Accessed: Sep. 03, 2021. [Online]. Available: http://arxiv.org/abs/1706.03762

[16] E. Agirre *et al.*, "SemEval-2016 Task 1: Semantic Textual Similarity, Monolingual and Cross-Lingual Evaluation," in *Proceedings of the 10th International Workshop on Semantic Evaluation (SemEval-2016)*, San Diego, California: Association for Computational Linguistics, Jun. 2016, pp. 497–511. doi: 10.18653/v1/S16-1081.